

# AI Engineering (Week 1)

ai26

## Topic 02 : Motivating Example

### Part 02 : Introduction to Data Operations

Preparation

Data Handling

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Building Models

Autumn Semester, 2026

#### Outline

- Characteristics of data sets
- Operations on tabular data

Wrap up

## AI Engineering (Week 2)

Introduction



Motivating Example

Preparation

Data Handling

Exploring Data 1

Exploring Data 2

Building Models

Prediction

Regression  
1

Classification  
1

Regression  
2

Classification  
2

Clustering

Wrap up

# Outline

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- |                         |    |
|-------------------------|----|
| 1. Introduction         | 3  |
| 2. The auto-mpg dataset | 6  |
| 3. Summary              | 14 |

# Data sources

| Type       | Format                | Example                   | DBMS                          | Language                     | Readiness for ML     |
|------------|-----------------------|---------------------------|-------------------------------|------------------------------|----------------------|
| Relational | Table                 | Transactions              | MySQL,<br>Postgresql,<br>...  | SQL                          | Maps to dataframe    |
| Flat       | Key + Value           | Caches                    | Redis, mem-<br>cached, ...    | DBMS-Specific                | Not rich enough      |
| Document   | Serialised<br>objects | Tweets                    | Mongodb,<br>Cassandra.<br>... | MQL, CQL                     | Too rich             |
| Graph      | Nodes and<br>edges    | Social rela-<br>tionships | Neo4j,<br>Dgraph, ...         | Gremlin, Cypher,<br>DQL, ... | Specialised analyses |
| Columnar   | DataSet               | Logs                      | HBase,<br>Spark<br>DataSet    | Hive QL, Spark SQL           | Maps to dataframe    |

Generally, **rich** flat data representations are best suited to machine learning

# Preparing data

Data Preparation is the first step in data mining

In practice, data can be

- structured or unstructured,
- consolidated or scattered,
- consistent or inconsistent,
- clean or with error.

ML prefers structured, consolidated, consistent data, as clean as possible.

The `auto_mpg.csv` dataset already has these characteristics.

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- |                         |    |
|-------------------------|----|
| 1. Introduction         | 3  |
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# The auto-mpg dataset

| 1  | mpg | cylinders | displacement | horsepower | weight | acceleration | model year | origin | car name                  |
|----|-----|-----------|--------------|------------|--------|--------------|------------|--------|---------------------------|
| 2  | 18  | 8         | 307          | 130        | 3504   | 12           | 70         | 1      | chevrolet chevelle malibu |
| 3  | 15  | 8         | 350          | 165        | 3693   | 11.5         | 70         | 1      | buick skylark 320         |
| 4  | 18  | 8         | 318          | 150        | 3436   | 11           | 70         | 1      | plymouth satellite        |
| 5  | 16  | 8         | 304          | 150        | 3433   | 12           | 70         | 1      | amc rebel sst             |
| 6  | 17  | 8         | 302          | 140        | 3449   | 10.5         | 70         | 1      | ford torino               |
| 7  | 15  | 8         | 429          | 198        | 4341   | 10           | 70         | 1      | ford galaxie 500          |
| 8  | 14  | 8         | 454          | 220        | 4354   | 9            | 70         | 1      | chevrolet impala          |
| 9  | 14  | 8         | 440          | 215        | 4312   | 8.5          | 70         | 1      | plymouth fury iii         |
| 10 | 14  | 8         | 455          | 225        | 4425   | 10           | 70         | 1      | pontiac catalina          |
| 11 | 15  | 8         | 390          | 190        | 3850   | 8.5          | 70         | 1      | amc ambassador dpl        |
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- ③ Each column stores a *variable*. Other terms include attributes and targets.
- ④ Machine learning uses combinations of these columns to build models.

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## Learning

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## Applied to auto-mpg

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- What other learning can be derived from the chosen column collection?

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- ... We can use *selection* (also known as *restriction*) - but how do we choose which rows to keep?

## Example

```
SELECT displacement, horsepower, weight, mpg  
FROM auto_mpg  
WHERE horsepower > 79;
```

- SELECT clause: projection (restricting columns: column sufficiency)
- WHERE clause: selection (restricting rows: row sufficiency)

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  - 1 Sampling - reduce/remove row duplication
  - 2 Banding - reduce the cardinality of a column
  - 3 Grouped Aggregation - roll up by level, aggregating as needed

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- Number of rows to keep in the sample is a compromise
- Can reduce runtime while allowing estimates of the uncertainty in a predictive model
- However, an aggregated column might be a better choice

```
-- Return a random sample of 3 Japanese cars  
SELECT *  
FROM AutoMpg  
WHERE originID = 3  
ORDER BY RANDOM()  
LIMIT 3;
```

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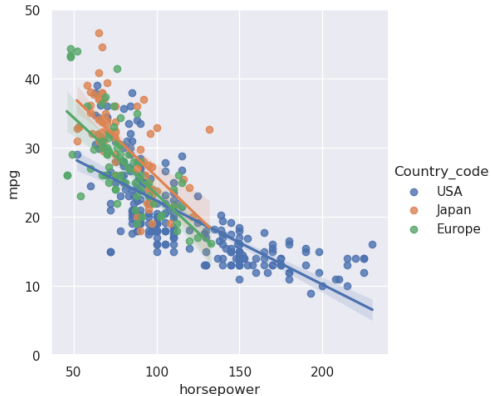
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## A grouped aggregation changes the effective key structure

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`COUNT(DISTINCT ...)`
- ... Take a set of values, compute an aggregate value
- Sets can be partitioned by grouping variable, aggregate applied to each partition
- ... Example: average mpg per country of manufacture

```
SELECT originID, AVG(mpg)
FROM AutoMpg
GROUP BY originID;
```

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# Summary

- Semantically rich, flat data is preferred for machine learning
- Ideally, this data would also be structured, consolidated, consistent and clean
- Several data operations were described, using the AutoMpg dataset as an example data source
  - Projection
  - Selection
  - Summarising: Sampling, Banding, Grouped Aggregation

Your task is to apply this to datasets using the python toolchain.